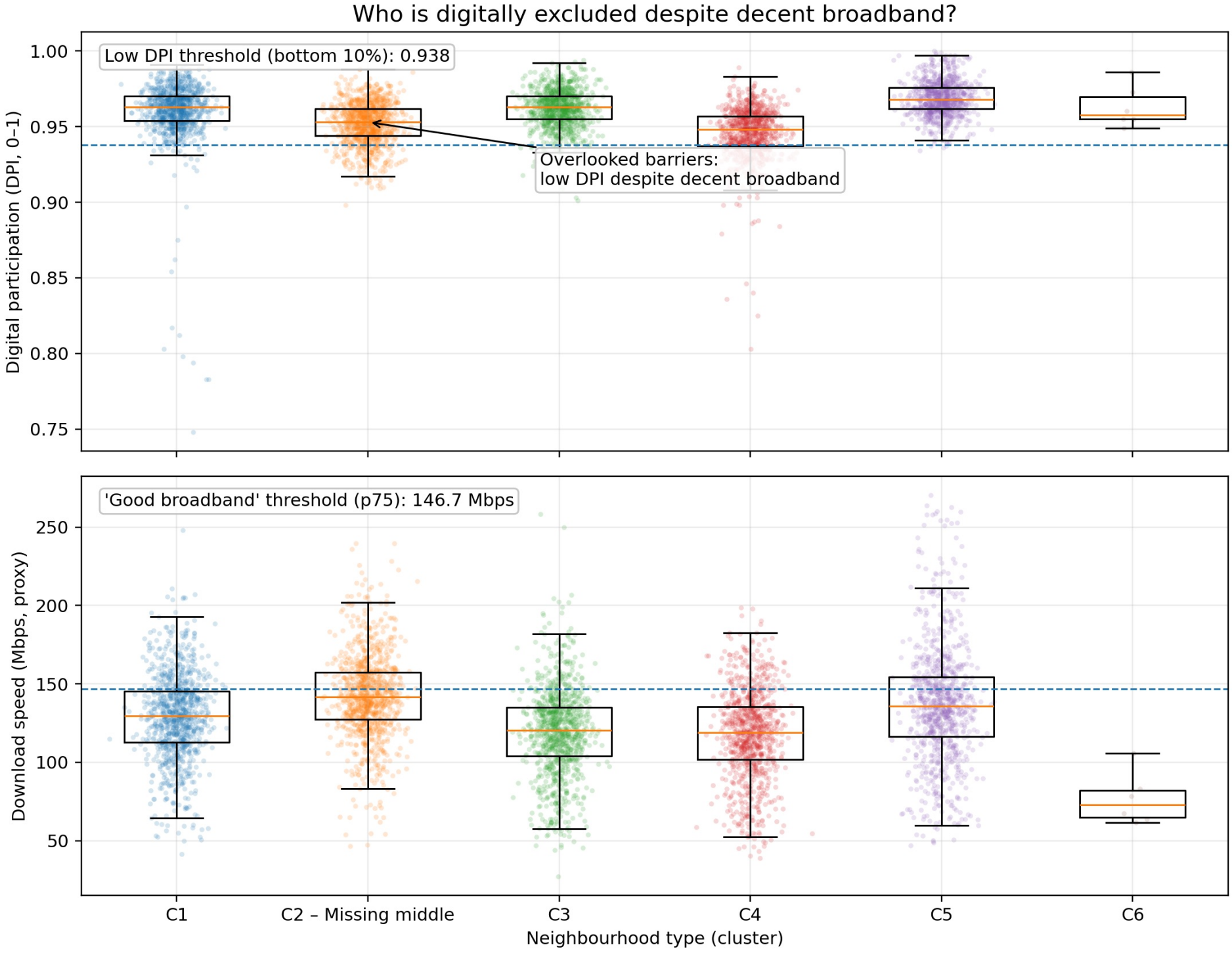
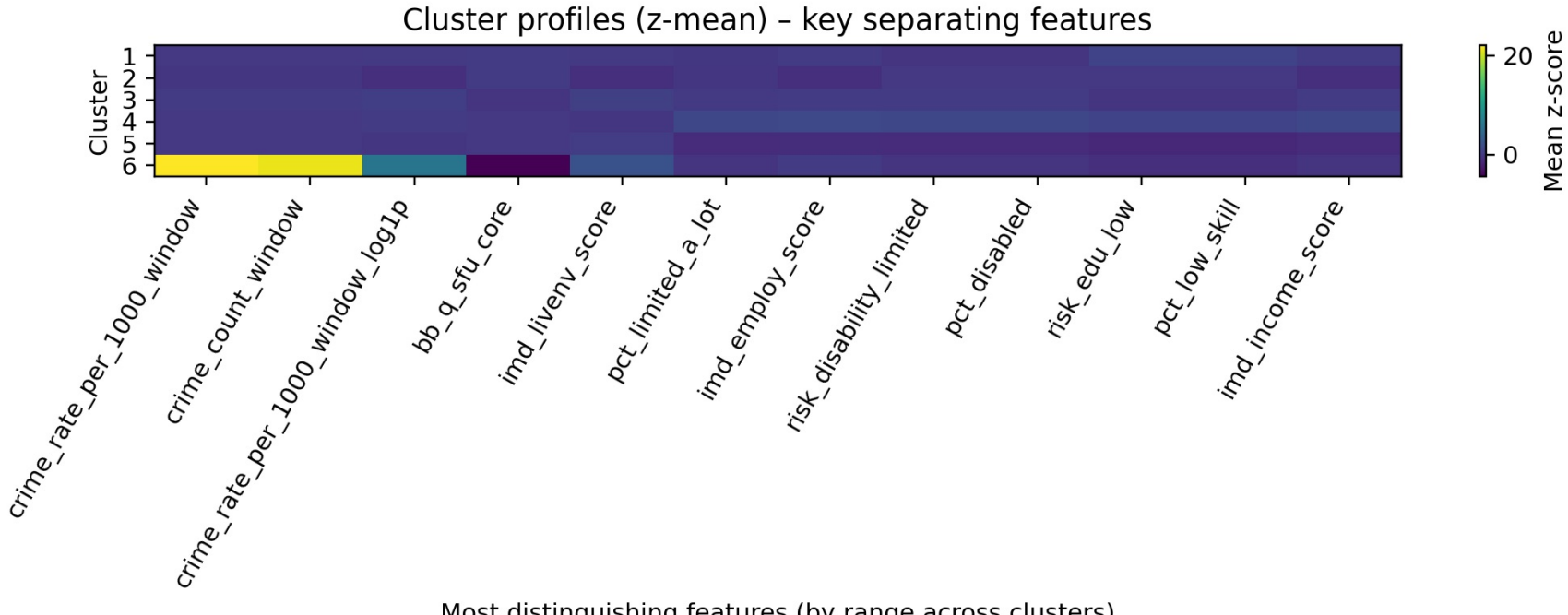
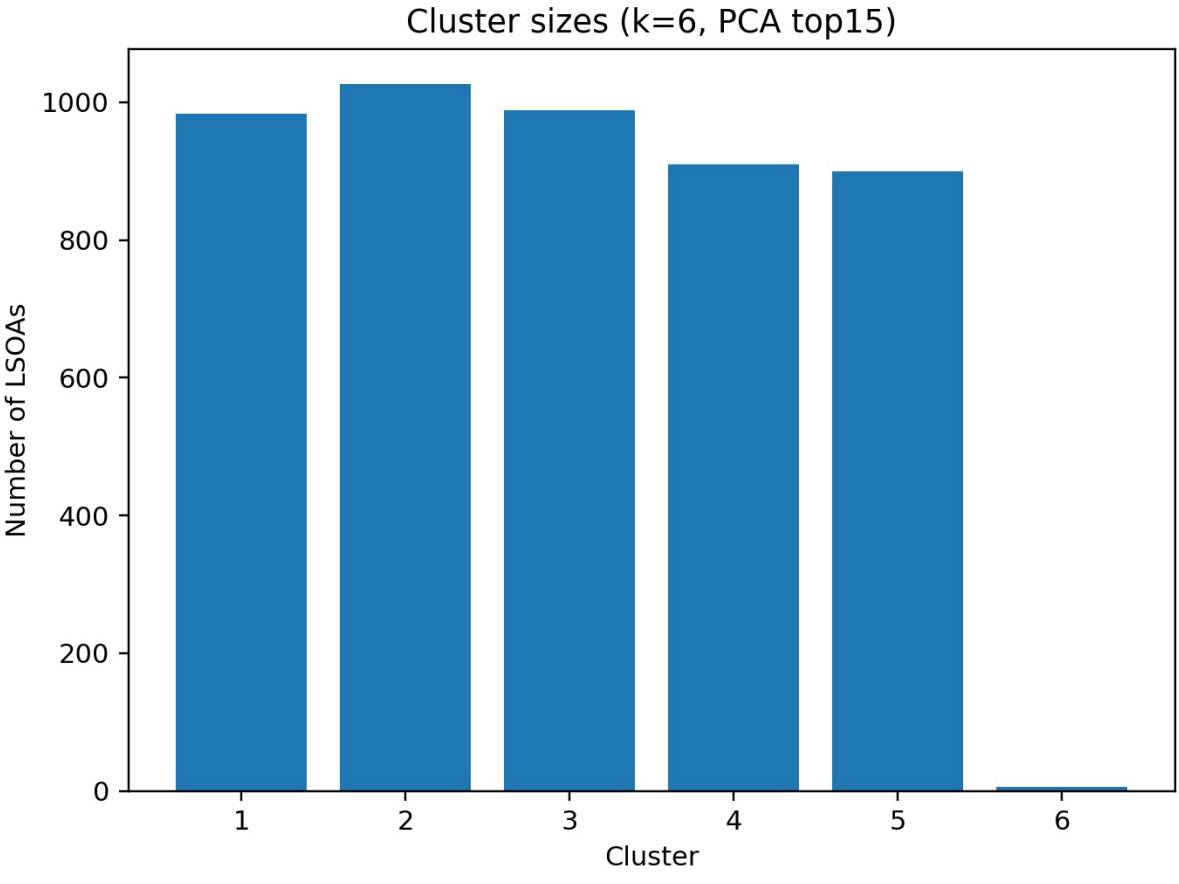
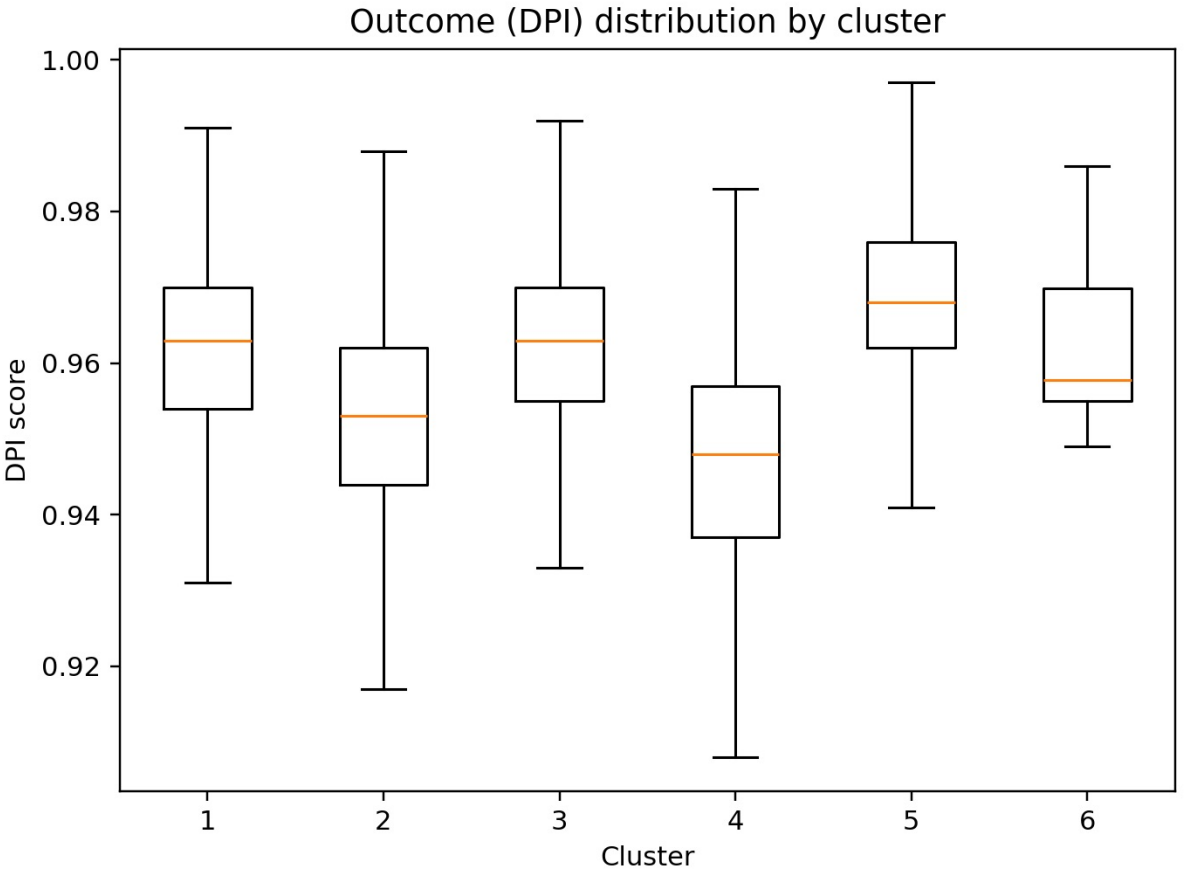

Who is digitally excluded despite decent broadband?
Low DPI threshold (bottom 10%): 0.938
Overlooked barriers:
low DPI despite decent broadband
Digital participation (DPI, 0–1)
1.00
0.95
0.90
0.85
0.80
0.75
'Good broadband' threshold (p75): 146.7 Mbps
Download speed (Mbps, proxy)
250
200
150
100
50
C1
C2 – Missing middle
C3
C4
C5
C6
Neighbourhood type (cluster)

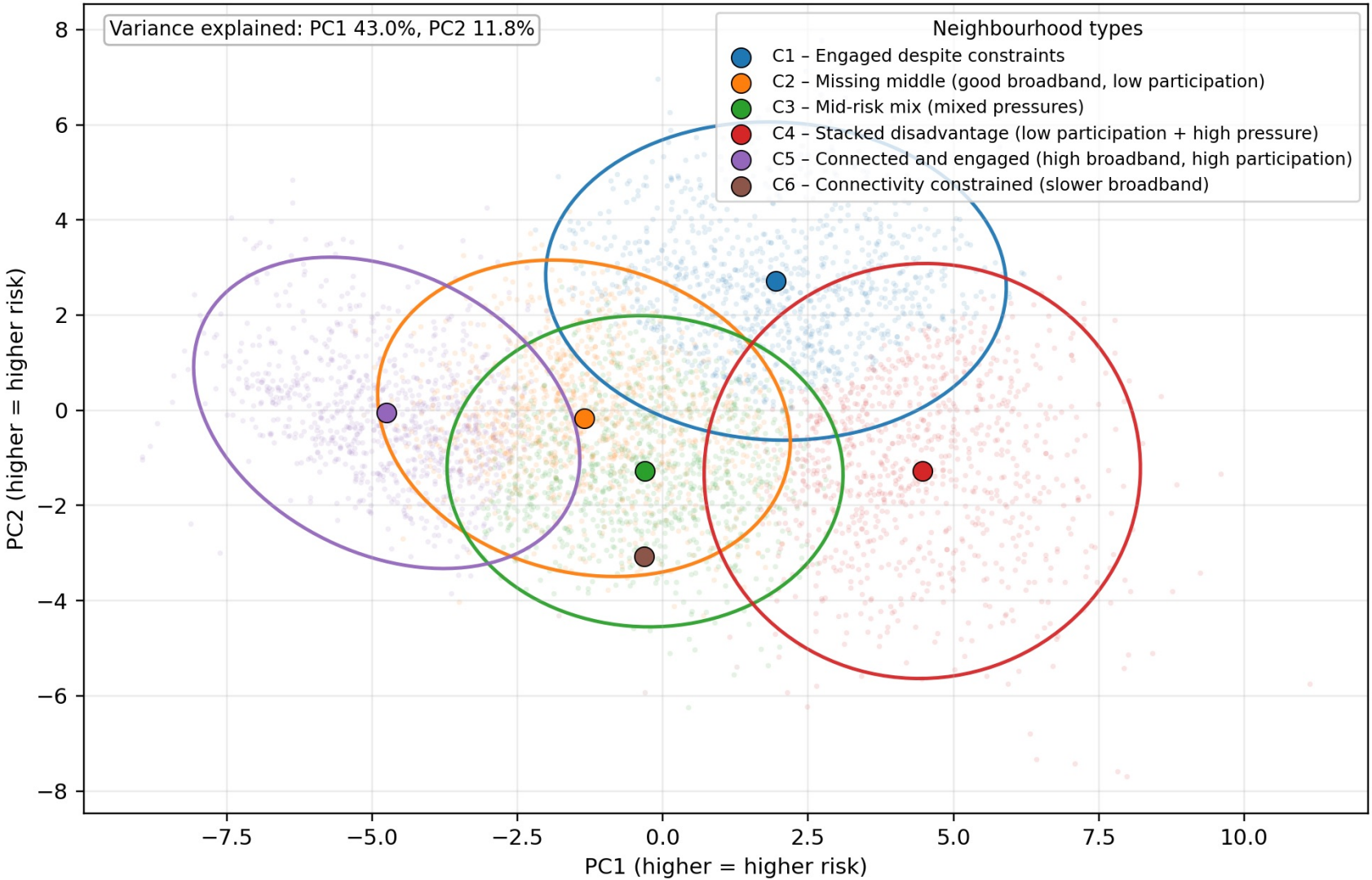

Cluster profiles (z-mean) – key separating features
Cluster
1
2
3
4
5
6
crime_rate_per_1000_window
crime_count_window
crime_rate_per_1000_window_log1p
bb_q_sfu_core
imd_livenv_score
pct_limited_a_lot
imd_employ_score
risk_disability_limited
pct_disabled
risk_edu_low
pct_low_skill
imd_income_score
Most distinguishing features (by range across clusters)
20
0
Mean z-score

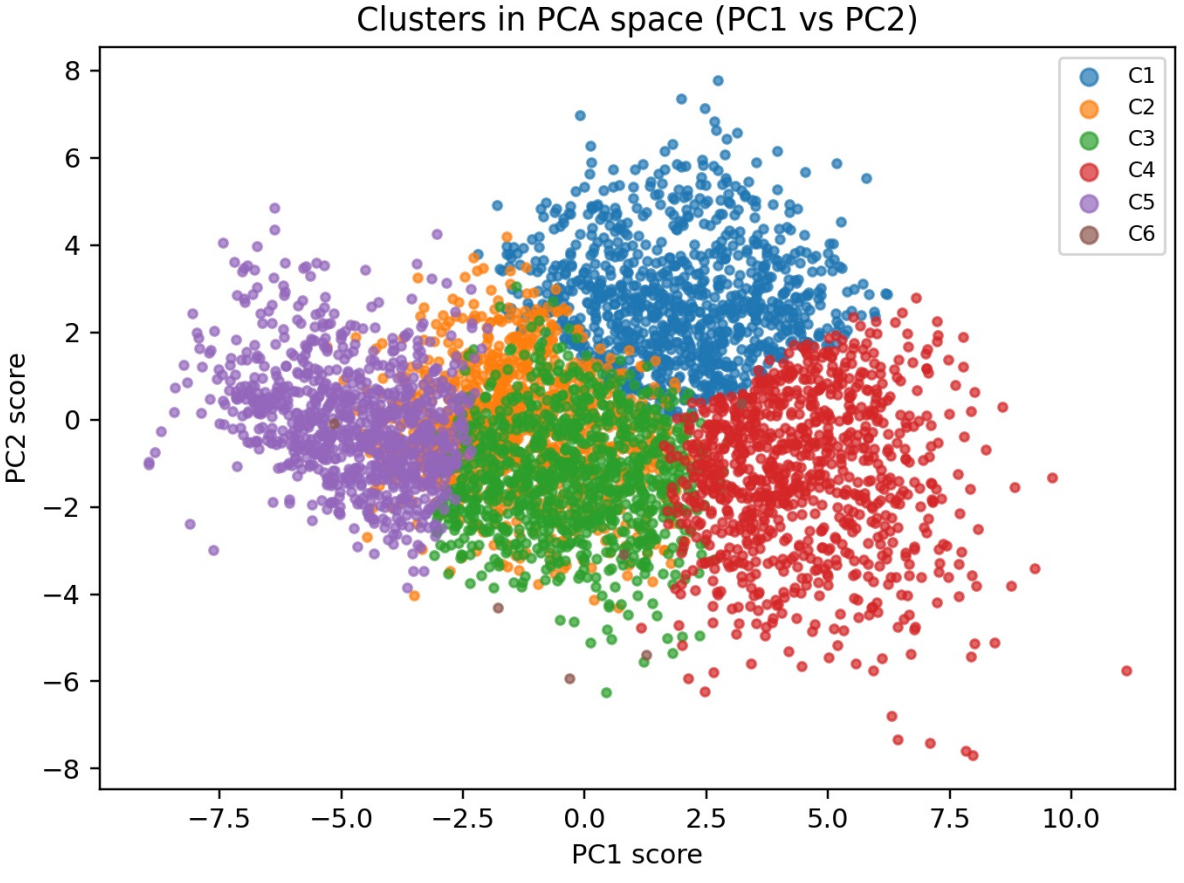

Cluster sizes (k=6, PCA top15)
Number of LSOAs
0
200
400
600
800
1000
1
2
3
4
5
6
Cluster

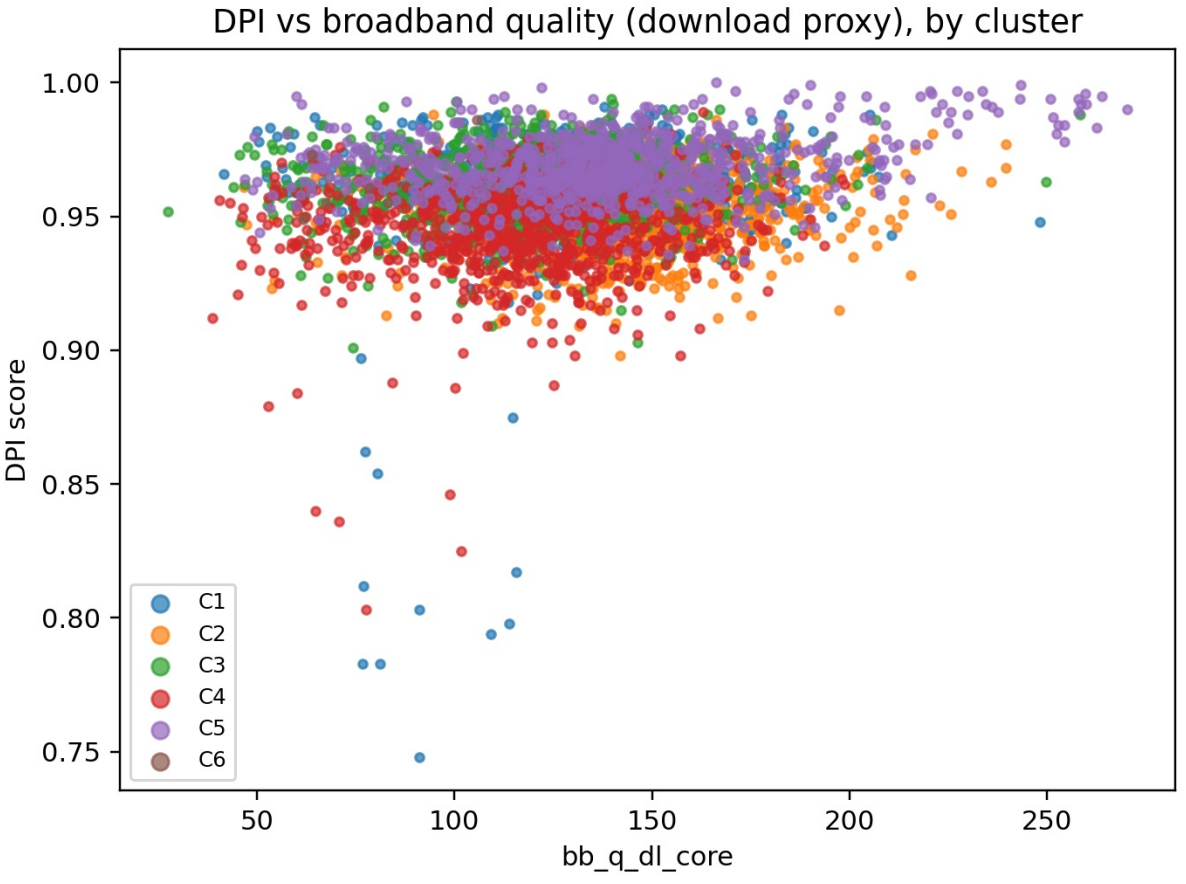

Outcome (DPI) distribution by cluster
1.00
0.98
0.96
0.94
0.92
DPI score
1
2
3
4
5
6
Cluster

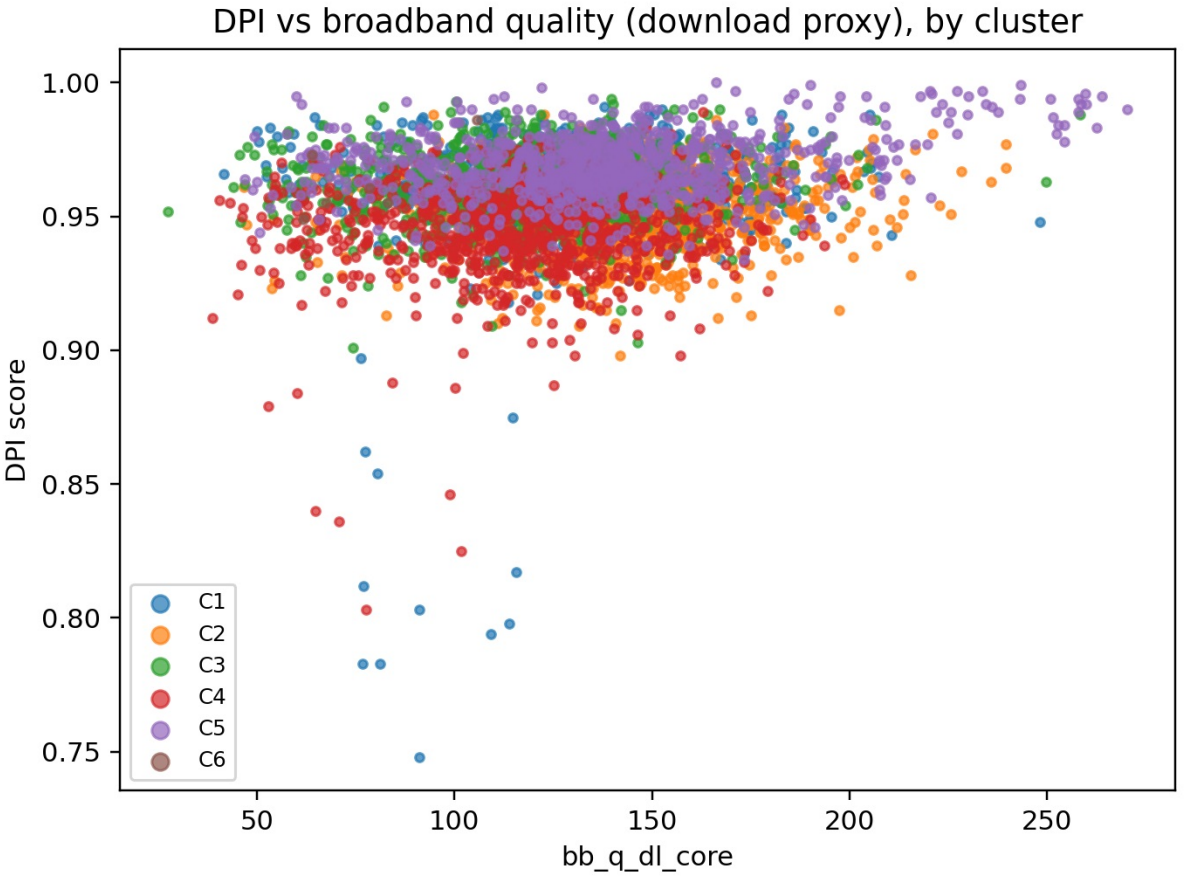
Six Neighbourhood Types Emerge from Multivariate Patterns

Clusters Shown in PCA Space (PC1 vs PC2)




Clusters in PCA space (PC1 vs PC2)
C1
C2
C3
C4
C5
C6
PC2 score
PC1 score
8
6
4
2
0
−2
−4
−6
−8
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0


DPI vs broadband quality (download proxy), by cluster
DPI score
bb_q_dl_core
1.00
0.95
0.90
0.85
0.80
0.75
50
100
150
200
250
C1
C2
C3
C4
C5
C6


DPI vs broadband quality (download proxy), by cluster
DPI score
bb_q_dl_core
1.00
0.95
0.90
0.85
0.80
0.75
50
100
150
200
250
C1
C2
C3
C4
C5
C6